CASE REPORT

Granuloma annulare and juvenile idiopathic arthritis: A unique case of coexistence

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ABSTRACT

Granuloma annulare (GA) is a chronic inflammatory skin condition with unclear pathophysiology, potentially related to immune dysregulations. It is associated with various systemic conditions, however, its connection to juvenile idiopathic arthritis (JIA) remains unclear. Here, we present a 14-year-old girl with oligoarticular juvenile idiopathic arthritis (JIA) and a two-year history of recurrent, mildly pruritic, symmetrical ankle lesions that were intermittent and self-resolving. A punch biopsy confirmed GA, and the patient was treated with topical steroids. This case highlights a unique association between granuloma annulare and juvenile idiopathic arthritis. (www.actabiomedica.it)

Key words: granuloma annulare, juvenile idiopathic arthritis, autoimmune diseases, inflammatory skin disorders, pediatric dermatology, immunopathogenesis.

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Introduction

Granuloma annulare (GA) is a chronic, idiopathic inflammatory disease characterized by granulomatous infiltration, leading to the development of annular plaques and papules, mainly affecting acral regions (1). The underlying pathophysiology of granuloma annulare remains largely unclear. Potential mechanisms include immune-mediated chronic microvascular injury and dysregulation of cell-mediated immunity. (2). Juvenile idiopathic arthritis (JIA) is a term for arthritis disorders of unknown etiology, persisting for more than six weeks and presenting before the age of 16. It is the most prevalent chronic inflammatory rheumatic disease in children (3). Cutaneous manifestations are commonly observed in systemic and polyarticular juvenile idiopathic arthritis (JIA). Systemic JIA, also known as adolescent-onset Still's disease, accounts for 5–15% of JIA cases. Skin involvement occurs in approximately 90% of affected individuals, typically presenting as well-defined, transient, salmon-pink macular or urticarial rashes distributed across the trunk, face, legs, or arms (4). To date, there was no evidence of an association between GA and JIA in the literature. Therefore, we present the case of a 14-year-old girl with a known history of JIA who subsequently developed granuloma annulare over her bilateral legs.

Case report

A 14-year-old girl with a medical history of oligoarticular juvenile idiopathic arthritis (JIA) for the past eight years presented to the dermatology clinic with a two-year history of recurrent, mildly itchy skin lesions over the bilateral ankles. The lesions appeared intermittently, lasting one to two days before resolving spontaneously. They were symmetrical but remained stable in size and number, and were not associated with pain, bleeding, or discharge. The patient denied any history of insect bites, trauma, recent vaccinations, travel, fever, or exposure to sick individuals. Her family history was unremarkable for similar skin conditions. She had been regularly followed up in the rheumatology clinic for JIA, experiencing multiple intermittent episodes of active arthritis, primarily affecting both knees. These episodes were characterized by pain and swelling, particularly after prolonged sitting, but responded well to intra-articular corticosteroid injections and short courses of naproxen (500 mg twice daily). Her last arthritis flare occurred three months before presentation, with no active joint symptoms at the time of evaluation. She denied morning stiffness, limping, or limitations in daily activities. On physical examination, a well-defined, light brown-red patch was observed over the dorsal aspect of bilateral ankles (Figure 1). No erythema, induration, scaling, or ulceration was noted, and there were no similar lesions elsewhere on the body. Systemic examination revealed no signs of active arthritis, lymphadenopathy, or organomegaly. Given the clinical presentation, the initial differential diagnoses included granuloma annulare (GA), erythema annulare centrifugum, interstitial granulomatous dermatitis, fixed drug eruption, and sarcoidosis. A 3 mm punch biopsy was performed on the left dorsal aspect of the ankle, which revealed histopathological findings consistent with granuloma annulare (Figure 2). Laboratory evaluation demonstrated normal inflammatory markers, including a low C-reactive protein level and erythrocyte sedimentation rate. These was no recent radiographic imaging of the knees had been performed. However, joint ultrasound revealed no evidence of active synovitis. At her most recent rheumatology follow-up, the patient's JIA was considered to be in clinical remission. Based on the clinical and microscopic evaluation, a diagnosis of GA was confirmed. The patient was started on topical mometasone furoate twice daily for one month, with complete resolution. She remained free of cutaneous recurrence during an approximate follow-up period of several months, with continued routine follow-up planned in both dermatology and rheumatology clinics to monitor for potential recurrence of granuloma annulare or reactivation of juvenile idiopathic arthritis.

Discussion

Granuloma annulare (GA) is a benign chronic inflammatory skin condition with unclear pathophysiology, possibly linked to immune dysregulations and microvascular injury (5). Juvenile idiopathic

Figure 1. Annular light brown non-scaly patches over left dorsal ankle.

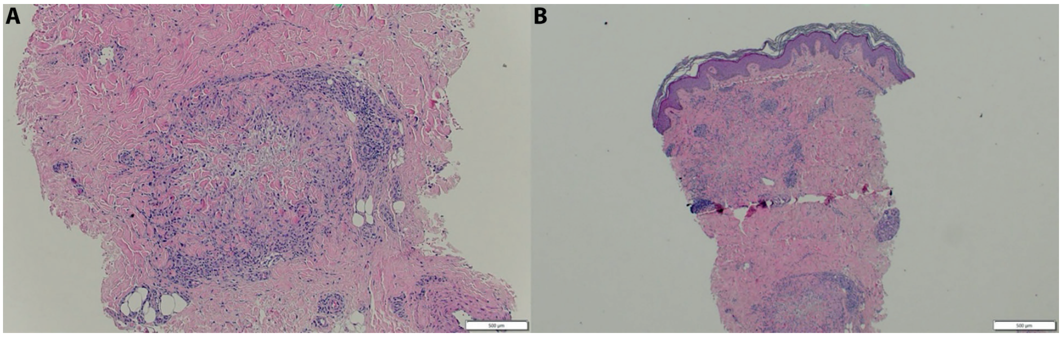

Figure 2. A) H&E section (10x) and B) (5x) reveal a well-circumscribed dermal granulomatous infiltrate with palisading histiocytes surrounding degenerated (necrobiotic) collagen, consistent with granuloma annulare. The surrounding dermis exhibits mild perivascular lymphocytic infiltration.

arthritis (JIA) is a collective term for all chronic forms of arthritis in children, impacting not only the joints but also extra-articular structures such as the eyes, skin, and internal organs. It is characterized by arthritis of unknown etiology that manifests before the age of 16 and persists for a minimum duration of six weeks (6). Both juvenile idiopathic arthritis (JIA) and granuloma annulare (GA) are immune-mediated inflammatory diseases marked by persistent inflammation and dysregulated T-cell responses. Though lesional tissues in both disorders show a preponderance of infiltrating $CD4^{+}$ T cells, JIA is largely driven by Th1 polarization with additional involvement of Th17 pathways, while GA is linked to a mixed Th1 and Th2 cytokine profile. Both conditions have been shown to have elevated levels of pro-inflammatory mediators, such as interferon-gamma (IFN-γ) and tumor necrosis factor-alpha (TNF-α), suggesting possible overlap in therapy targets like TNF-directed biologic therapies. Furthermore, macrophages and dendritic cells play integral roles in the etiopathogenesis of both diseases by sustaining and amplifying inflammatory signaling networks. Collectively, these shared immunological features suggest overlapping T-cell–mediated inflammatory mechanisms manifesting in distinct clinical (7,8). Several studies have found that patients with juvenile idiopathic arthritis (JIA) are at increased risk of developing other autoimmune conditions, particularly autoimmune thyroid disease, type 1 diabetes mellitus, rheumatoid arthritis, systemic lupus erythematosus, ankylosing spondylitis, psoriatic arthritis, and Sjögren's syndrome (9). Similarly, a large study was conducted that highlights that GA is associated with systemic conditions like diabetes and autoimmune diseases including rheumatoid arthritis, systemic lupus erythematosus, and alopecia areata among others, however, it was on the adult population (2). Another large-scale pediatric study showed a significant association between (GA) and Type I diabetes mellitus, hematologic malignancies, and atopy. Additionally, a recent case series further emphasized diabetes mellitus, thyroid disease, and hyperlipidemia as comorbidities of pediatric GA, which was observed in small numbers (10,11). Importantly, in the present case, GA developed while the patient's JIA was clinically inactive, with normal inflammatory markers and no ultrasonographic evidence of active synovitis, arguing against a direct temporal relationship between systemic inflammatory activity and the cutaneous manifestation. Moreover, our patient denied any history of insect bites, trauma, vaccinations, travel, fever, or exposure to sick individuals. However, several studies have implicated viral infection as an initiating factor in GA, including Epstein-Barr virus, HIV, varicella zoster virus, and SARS-CoV-2, yet no such history was present in this case (7,12). The role of medications, particularly non-steroidal anti-inflammatory drugs (NSAIDs), Allopurinol, and corticosteroids, has also been debated, with some reports suggesting that these drugs may either exacerbate or ameliorate GA (13). Since our patient received intermittent naproxen and intra-articular corticosteroids, it's unclear whether these treatments influenced GA, warranting further research into their effects on GA.

In this case, the patient presented with localized GA in the setting of oligoarticular JIA and was managed conservatively with topical mometasone furoate, resulting in significant clinical improvement.

Conclusion

This study reports an unusual and previously unknown coexistence of granuloma annulare and juvenile idiopathic arthritis. While this data suggests the possibility of similar immunopathogenic pathways, it is still an isolated finding and may be coincidental. More case reports and mechanistic papers are required before any clinical association can be established.

Ethic approval: All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. This is an observational study. For this type of study, formal consent is not required because no personal data was contained, and there is no concern about identifying information.

Conflict of interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

Authors' contribution: All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by NA and SA. The first draft of the manuscript was written by NA., and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript

Declaration on the use of AI: None.

Consent for publication: The patient in this manuscript has given written informed consent.

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